SESSION 1

# Evolution of IT Operations & AIOps

*From Reactive Monitoring to AI-Driven Intelligence*

- Limitations of Traditional Monitoring
- Digital Operations vs AIOps
- Business Drivers & ROI
- AIOps Maturity Models

Ram N Sangwan

# Session Overview

*What we will cover in this chapter*

**01**

### History of IT Operations

Evolution from mainframes to cloud-native environments

**02**

### Limitations of Traditional Monitoring

Alert fatigue, siloed tools, reactive posture

**03**

### What is AIOps?

Gartner definition, key components, and capabilities

**04**

### Digital Operations vs AIOps

How AIOps extends digital operations

**05**

### Business Drivers & ROI

Key metrics, cost savings, and strategic value

**06**

### AIOps Maturity Models

5-level framework and your organization's journey

# History of IT Operations

*The journey from manual processes to intelligent automation*

**1960s–70s**

**Mainframe Era**

Centralized computing, batch processing, manual operations

**1980s–90s**

**Client-Server**

Distributed systems, early SNMP monitoring, help desks

**2000s**

**Web & ITSM***

ITIL processes, APM tools, NOC growth, service desks

**2010s**

**Cloud & DevOps**

Agile, containers, microservices, observability tools

**2020s+**

**AIOps & AI-Driven**

ML, automation, predictive analytics, self-healing systems

1960s 1980s 2000s 2010s 2020s+

*Key Insight: Modern IT environments have grown too complex for manual human management alone.*



***Information Technology Service Management**

# The Complexity Problem

*Why traditional monitoring is breaking down*

**87%**

of enterprises run hybrid/multi-cloud

**10x**

more telemetry from microservices vs monoliths

**3.5M**

IT workforce gap (unfilled positions 2025)

**99%**

event volume reduction possible with AIOps

# Limitations of Traditional Monitoring

*The challenges driving the need for AIOps*

### Alert Fatigue

Static thresholds trigger thousands of low-quality alerts, desensitizing on-call teams and hiding critical signals in noise.

### Siloed Data & Tools

Separate dashboards for CPU, logs, tickets, and network prevent holistic visibility and correlation across systems.

### Reactive Posture

Issues are detected only after users experience impact, leading to poor SLA performance and customer dissatisfaction.

### Manual Correlation

Engineers spend hours correlating events across systems during incidents, slowing mean-time-to-resolution (MTTR).

### Static Rules

Rule-based alerting cannot adapt to dynamic cloud environments, causing both false positives and missed anomalies.

### Scaling Challenges

As infrastructure grows exponentially, human teams cannot maintain pace with the volume and velocity of operational data.

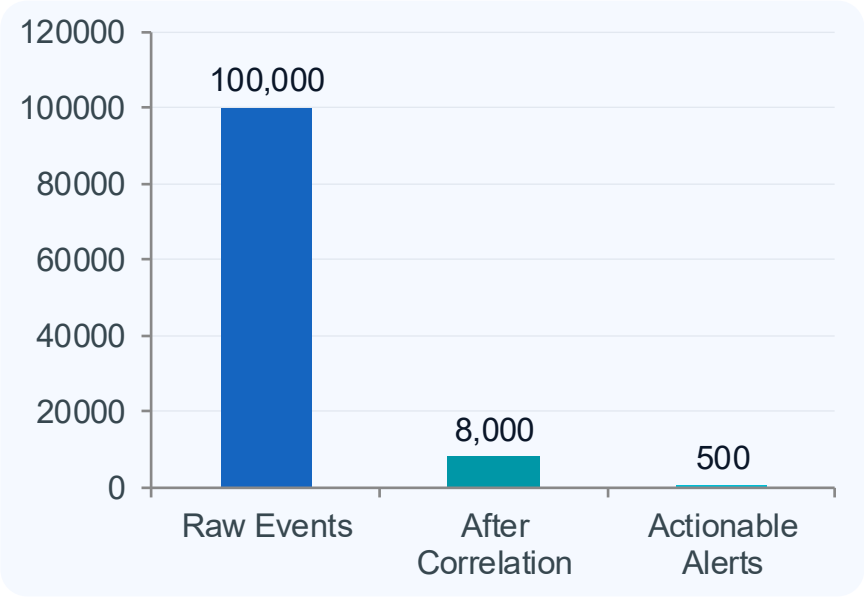
# Alert Fatigue: A Critical Challenge

*Traditional monitoring generates overwhelming noise*

## The Alert Noise Problem

- On-call engineers receive hundreds to thousands of alerts daily
- Up to 60% of alerts in traditional systems are noise or duplicates
- Alert storms during incidents create cognitive overload
- Teams develop 'alarm blindness' — ignoring critical signals
- Static thresholds fail in dynamic cloud environments
- No correlation between related events across tools

**Event Volume Reduction**

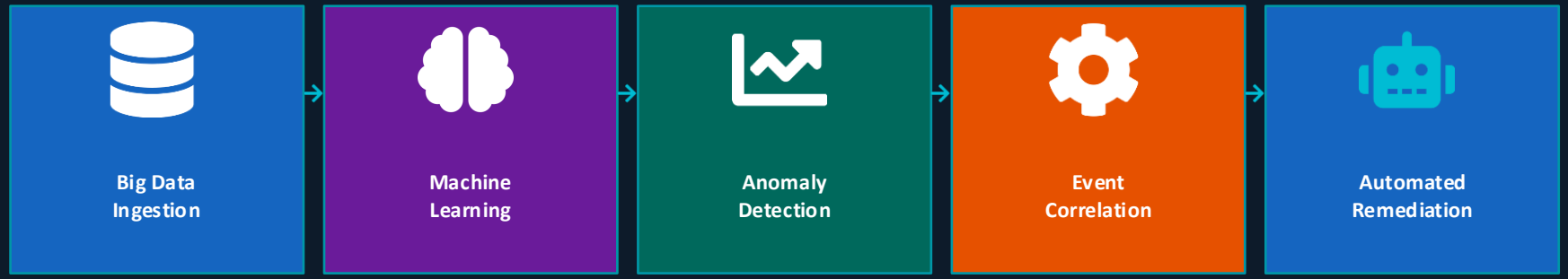



*Source: ServiceNow Predictive AIOps – up to 99% event volume reduction*

# What is AIOps?

*"AIOps combines big data and machine learning to automate IT operations processes, including event correlation, anomaly detection and causality determination."*

*— Gartner Definition*




**AIOps = AI/ML + Big Data + Automation applied to IT Operations**

# How AIOps Works

*The data pipeline from ingestion to intelligent action*

**1**

**Data Ingestion**

Logs, metrics, events, traces from all sources

**2**

**Data Processing**

Normalization, enrichment, correlation

**3**

**ML Analysis**

Anomaly detection, pattern recognition, baseline learning

**4**

**Insight Generation**

Root cause analysis, impact assessment, predictions

**5**

**Automated Action**

Remediation, ticket creation, self-healing

**Data Sources:**

- Monitoring Tools
- APM Systems
- Cloud Services
- ITSM / CMDB
- Network Telemetry
- Log Aggregators

# Digital Operations vs AIOps

*Understanding the relationship and key differences*

## Digital Operations

- Broad framework for technology-driven business operations
- Encompasses DevOps, CloudOps, SecOps, FinOps
- Focuses on end-to-end digital service delivery
- Uses automation and data analytics broadly
- Organizational capability and culture
- Manual intervention still common for complex issues

**AIOps ⊂ DigOps**

## AIOps

Specific AI/ML layer applied to IT operations data

Focused on monitoring, events, and incident management

- Real-time analysis of operational telemetry
- Autonomous anomaly detection and root cause analysis
- Technology capability powered by ML models
- Enables self-healing and predictive prevention

*AIOps is a core intelligence engine that powers and accelerates Digital Operations.*

# Traditional IT Ops vs AIOps

*A side-by-side comparison of operating models*

| Dimension | Traditional IT Operations | AIOps-Powered Operations |
|---|---|---|
| **Detection** | Reactive — users report issues | Proactive — anomalies detected pre-impact |
| **Alerting** | Static threshold rules, high noise | ML-based dynamic baselines, contextual alerts |
| **Root Cause** | Manual investigation, hours/days | Automated correlation, minutes |
| **Response** | Human-driven runbook execution | Automated remediation & self-healing |
| **Scalability** | Scales poorly with growth | Scales elastically with environment |
| **Data** | Siloed per tool | Unified data plane — logs, metrics, traces |
| **MTTR** | Hours to days | Minutes to seconds |

# Business Drivers for AIOps

*Why organizations are accelerating AIOps adoption*

### Cloud Complexity

87% of enterprises now operate hybrid/multi-cloud environments, generating volumes of telemetry that outpace manual management.

### Digital Transformation

Every major enterprise is accelerating digitization, requiring IT operations to support higher release velocity and reliability.

### Cost of Downtime

$2M+ per hour lost in financial services from downtime — AIOps directly prevents revenue loss through proactive detection.

### Talent Shortage

3.5M unfilled IT/cybersecurity positions globally — AIOps multiplies the effectiveness of existing teams through automation.

### Customer Experience

Customers demand near-zero downtime. Slow resolution leads to churn, abandoned transactions, and brand damage.

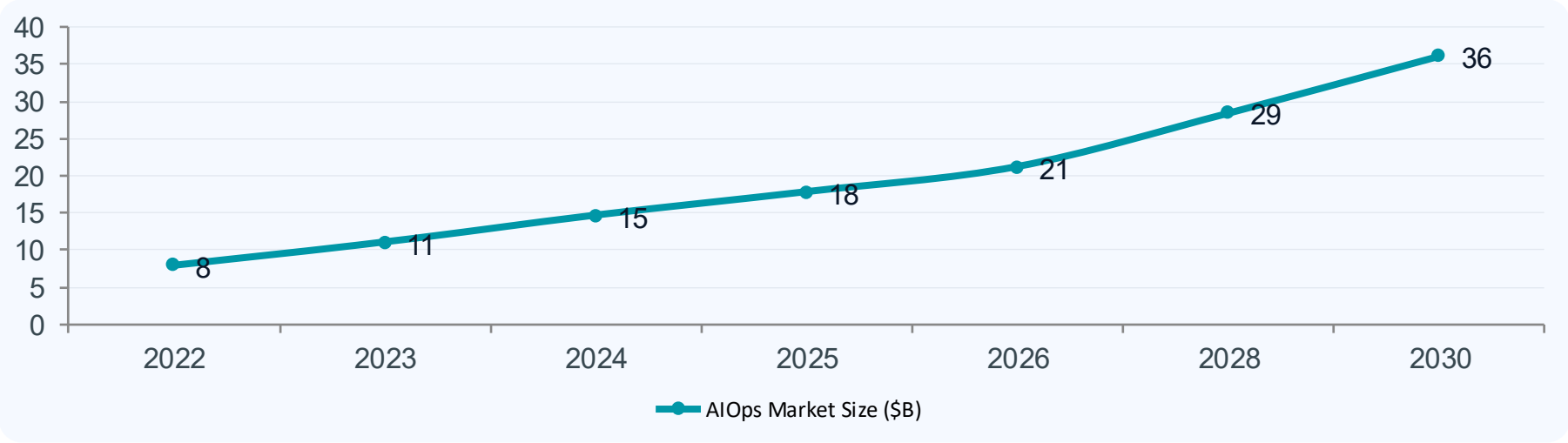
### Security & Compliance

AIOps enables continuous compliance monitoring and behavioral anomaly detection for proactive threat identification.

# AIOps Market Growth

*A rapidly expanding market driven by enterprise need*

| $14.6B | $36.1B | 15.2% | 54% |
|---|---|---|---|
| Market Size 2024 | Projected 2030 | CAGR 2025–2030 | Enterprise Adoption 2025 |




Source: Grand View Research AIOps Platform Market 2024–2030 | CloudEagle.ai AIOps Guide | https://www.grandviewresearch.com/industry-analysis/aiops-platform-market

# AIOps ROI & Business Value

*Quantifiable returns and strategic benefits*

**90%**

Reduction in average downtime costs (from $23.9M to $2.5M annually)

*ESG Research*

**99%**

Event volume reduction through intelligent correlation and noise filtering

*ServiceNow*

**30%**

Reduction in operational costs via hyperautomation (Gartner forecast)

*Gartner*

**79%**

Organizations reported reduced errors after adopting process automation

*IDC Research*

*Key Consideration: ROI varies — focused initial deployments on high-value use cases yield the fastest returns.*

Source: ESG Research via ScienceLogic | ServiceNow Predictive AIOps | Gartner Hyperautomation | https://sciencelogic.com/blog/2023-trends-in-aiops-observability-and-itops

# AIOps ROI Calculation Framework

*How to measure and justify AIOps investment*

## COSTS (Investment)

- Platform licensing & subscription
- Implementation & integration costs
- Training & skill development
- Managed services (if applicable)
- Ongoing model tuning & governance

## RETURNS (Benefits)

- Downtime reduction × hourly cost of outages
- MTTR improvement × incident frequency
- Engineer productivity gains (hrs saved × labor cost)
- Alert noise reduction (on-call hour savings)
- Automation of routine tasks (FTE equivalent)

**ROI (%) = [(Total Benefits – Total Costs) ÷ Total Costs] × 100 | Payback Period = Costs ÷ Annual Benefits**

Source: Mordor Intelligence AIOps Market | diffstudy.com AIOps Implementation Guide | https://www.mordorintelligence.com/industry-reports/aiops-market

# AIOps Key Use Cases

*Real-world applications delivering measurable value*

## 01 Incident Management

Automatic correlation of events, root cause analysis, and AI-guided remediation to reduce MTTR from hours to minutes.

## 02 Anomaly Detection

ML baselines dynamically detect unusual patterns in metrics, logs, and traces — including subtle degradations invisible to static rules.

## 03 Predictive Analytics

Forecast capacity bottlenecks, hardware failures, and performance issues before they impact services.

## 04 Change Risk Analysis

Assess impact of proposed changes by correlating historical change data with incidents to reduce change-related failures.

## 05 Digital Experience

Correlate technical metrics with business KPIs (conversion rates, revenue) to prioritize fixes by customer impact.

## 06 Security Operations

Apply ML to detect behavioral anomalies, correlate security events with operational data, and automate threat responses.

Source: IntelligentVisibility – 5 AIOps Enterprise Use Cases | https://intelligentvisibility.com/blog/5-aiops-enterprise-use-cases-roi-examples

# AIOps Maturity Models

*Understanding where your organization stands — and the roadmap to reach full AIOps maturity.*

## What is a Maturity Model?

A maturity model provides a structured framework for assessing an organization's current AIOps capabilities, identifying gaps, and building a roadmap for progressive enhancement. It helps IT leaders link operational capability with measurable business outcomes.

## Key Reference Frameworks:

| | |
|---|---|
| **TM Forum** | IG1322 AIOps Maturity Model (Approved Feb 2024) |
| **Gartner** | IT Operations Maturity Assessment Framework |
| **Virtana** | 5-Level AIOps Maturity Model (2025) |
| **Arion Research** | AIOps Maturity — Agentic Operations Framework |

# The 5 Levels of AIOps Maturity

*Progressive stages from reactive to autonomous operations*

| L1 Reactive | L2 Aware | L3 Proactive | L4 Adaptive | L5 Autonomous |
|---|---|---|---|---|
| Manual processes, siloed tools, reactive incident response. Ad hoc operations with minimal automation. | Basic monitoring integrated, manual data analysis. Improved visibility but still largely reactive. | Predictive analytics applied. Event correlation across tools. Anomaly detection reduces alert noise. | Automated remediation for known issues. AI-driven insights. Continuous learning from incidents. | Self-healing systems. Prescriptive AI. Full automation with ML feedback loops. Near-zero human intervention. |

← Increasing Automation, Intelligence & Business Value →

# Maturity Levels 1 & 2: Reactive & Aware

*The starting point for most organizations*

## Level 1: Reactive

**Characteristics:**

- Ad hoc manual processes
- Siloed data and operations
- Reactive to incidents after impact
- Basic monitoring tools only
- Manual alert interpretation

Capabilities: Basic alerting, manual data analysis, no automation

⚠ Challenges: High operational costs, frequent downtime, slow resolution

## Level 2: Aware

**Characteristics:**

- Integrated monitoring tools
- Log aggregation in place (Splunk, ELK)
- APM tools deployed (AppDynamics, New Relic)
- Some cross-team data sharing
- Manual correlation still dominant

Capabilities: Multiple data sources, deeper insights per component, improved visibility

⚠ Challenges: Still largely reactive — no unified view, alert storms common

# Maturity Levels 3 & 4: Proactive & Adaptive

*Where most AIOps investments target today*

## Level 3: Proactive

**Capabilities:**

- Predictive analytics applied to operational data
- Event correlation across multiple systems
- ML-based anomaly detection reduces noise
- Early warning signals prevent some outages
- Automated alerts for high-confidence issues

✓ Capabilities: Anomaly detection, signal processing, cross-tool correlation

📈 Business Gain: Organizations see 40–60% reduction in manual investigation time

## Level 4: Adaptive

**Capabilities:**

- Automated remediation for known failure patterns
- Self-healing by auto-scaling/rerouting services
- AI-driven insights with contextual recommendations
- Continuous learning from incident history
- Integration with ITSM for auto-ticket creation

✓ Capabilities: Prescriptive recommendations, automated playbooks, ML feedback

📈 Business Gain: MTTR reduces from hours to minutes; on-call burden significantly reduced

# Level 5: Autonomous Operations

*The pinnacle of AIOps maturity — self-healing, prescriptive, and continuously learning systems*

### Self-Healing Systems

Automated detection, diagnosis, and remediation without human intervention. CI/CD integrated.

### Prescriptive Analytics

AI not only predicts issues — it recommends and implements optimal resolutions in real-time.

### ML Feedback Loops

Continuous learning from every incident improves model accuracy and reduces false positives over time.

### Agentic AI Operations

AI agents observe, recommend, act, and adapt — evolving from tools to autonomous operational co-pilots.

*Only a fraction of organizations have reached Level 5 — but it defines the target operating model for AIOps maturity.*

# AIOps Maturity Assessment Dimensions

*Key dimensions to evaluate your organization's maturity level*

| Dimension | L1 Reactive | L2 Aware | L3 Proactive | L4 Adaptive | L5 Autonomous |
|---|---|---|---|---|---|
| **Data Maturity** | Siloed | Integrated | Unified | Real-time | **Predictive** |
| **Automation Depth** | None | Basic | Partial | Advanced | **Full Auto** |
| **Human-AI Collab** | Manual | Assisted | Guided | Augmented | **Supervised** |
| **Event Management** | Alerts | Filtered | Correlated | Predicted | **Self-healed** |
| **Analytics Capability** | None | Descriptive | Predictive | Prescriptive | **Autonomous** |
| **Governance & AI** | None | Minimal | Defined | Managed | **Optimized** |

Source: TM Forum IG1322 AIOps Maturity Model | Arion Research AIOps Maturity | Info-Tech Research | https://www.tmforum.org | https://www.infotech.com

# AIOps Tools & Platform Landscape

*Leading platforms and vendors in 2025–2026*

## Dynatrace

Full-Stack Observability

AI-powered full-stack observability with automated root cause analysis (Davis AI)

## ServiceNow

ITSM + AIOps

Predictive AIOps for ITSM integration — event correlation, anomaly detection

## Splunk / Cisco

Security + ITOps

IT Service Intelligence — security & operations correlation (acquired by Cisco 2024, $28B)

## IBM Watson AIOps

Enterprise AI

Enterprise-grade event management and IT automation

## Datadog

Cloud Monitoring

LLM Observability (2025) for AI workloads + full-stack cloud monitoring

## BigPanda

AIOps Specialist

Advanced event correlation — cuts alert noise by up to 90%

*Top 5 vendors (Dynatrace, Splunk, Datadog, IBM, ServiceNow) control ~38% of global AIOps market revenue (2025)*

# AIOps Implementation Roadmap

*A practical phased approach to AIOps adoption*

## Phase 1
Foundation
(0–3 months)

- Audit monitoring tools & data sources
- Establish MTTR & alert volume baselines
- Assess CMDB & service mapping maturity
- Define business case & ROI targets

## Phase 2
Observe
(3–6 months)

- Deploy AIOps in observation mode
- Connect all data sources to platform
- Allow ML models to learn baselines
- Tune models to reduce false positives

## Phase 3
Correlate
(6–12 months)

- Enable event correlation & grouping
- Implement initial automated playbooks
- Integrate with ITSM for auto-ticketing
- Measure MTTR improvement progress

## Phase 4
Automate
(12–18 months)

- Expand automated remediation coverage
- Activate predictive analytics & forecasting
- Implement self-healing for known patterns
- Optimize based on continuous learning

## Phase 5
Optimize
(18+ months)

- Achieve full AIOps maturity (L4/L5)
- Continuous model improvement & retraining
- Align IT ops metrics to business KPIs
- Explore agentic AI for autonomous ops

*Key: Start small — focus first deployment on ONE high-value use case with clear baseline metrics for fastest ROI.*

# Session 1 — Key Takeaways

1. Traditional monitoring is broken — Alert fatigue, siloed tools, and reactive posture cannot scale with modern cloud environments.
2. AIOps = AI/ML + Big Data + Automation — Gartner defines it as the intelligent layer that automates IT ops through correlation, anomaly detection, and causality determination.
3. AIOps is a subset of Digital Operations — It serves as the intelligence engine that powers broader digital transformation initiatives.
4. Business ROI is real and measurable — 90% downtime cost reduction, 30% operational cost savings, 99% event noise reduction are documented outcomes.
5. The AIOps market is growing fast — From $14.6B in 2024 to $36.1B by 2030 (15.2% CAGR), driven by cloud complexity and talent gaps.
6. Maturity is a journey — Start from Level 1 (Reactive) and progressively build toward Level 5 (Autonomous). Focus on ONE high-value use case first.